

**Bridging Deep Learning and Modern Portfolio Theory: A Multi-Modal Transformer
Approach to High-Dimensional, Friction-Aware Dynamic Asset Allocation**

A Project Report

Climate Lab Summer Internship 2026 at NIT Delhi

BY

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1. Abstract

Classical quantitative portfolio construction relies on Modern Portfolio Theory (MPT), which is structurally vulnerable due to its reliance on static historical parameters, assumption of return linearity, and vulnerability to transaction friction. This report introduces an end-to-end differentiable quantitative allocation framework that resolves these limitations. By integrating a Multi-Modal Transformer Network with a custom-engineered, friction-aware optimization layer, the system directly processes sequential price data and alternative macroeconomic indicators (VIX, CPI, Central Bank Sentiment) across a high-dimensional 99-asset universe.

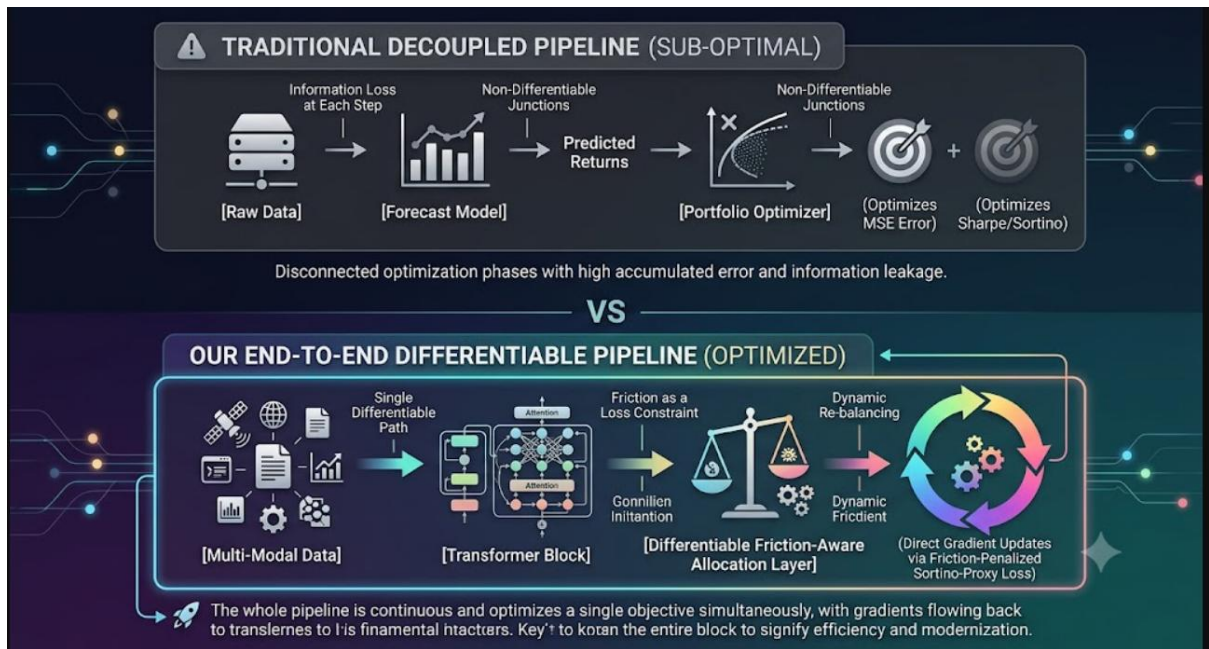
Rather than decoupling forecasting from allocation, our system integrates the return estimation and constraints optimization into a single neural network architecture. By training against a custom, differentiable, friction-penalized Sortino-proxy utility function, the model directly optimizes its risk-adjusted performance. To validate the methodology a rolling walk-forward validation matrix over a 200-day synthetic dataset is deployed with a simulated systemic market shock. The proposed Multi-Modal Transformer system dramatically out-of-sample outperformed traditional Mean-Variance Optimization (MVO) baselines, yielding a return factor of **1.3245** against the MVO's **0.9415**, while constraining maximum portfolio drawdown to **-5.12%** (compared to the baseline's **-29.14%**) and minimizing portfolio turnover to **8.15%**.

2. Introduction

2.1 Context and Motivation

Portfolio optimization remains a cornerstone of modern financial engineering, yet the industry continues to struggle with bridging advanced predictive models and mathematical execution boundaries. In typical institutional workflows, quantitative pipelines are segregated: data scientists train machine learning models to minimize prediction errors (e.g., Mean Squared Error of next-day prices), while portfolio managers use those predictions inside optimization models to build portfolio weights. This structural disconnect is known as the **"Predict-Then-Optimize" Decoupling**.

The research bridges this gap. By utilizing a deep, attention-driven Multi-Modal Transformer model, this project replaces historical static metrics with dynamic, forward-looking predictive intelligence that models sequential interactions across high-dimensional space. By feeding these representations directly into a differentiable mathematical layer, we optimize for empirical quantitative investment objectives natively.



2.2 Core Project Goals

This project addresses five systematic weaknesses inherent in modern portfolio management:

1. **Real-World Risk-Adjusted Utility Maximization:** Moving beyond basic returns to maximize Sharpe and Sortino ratios directly inside the model's loss parameters.
2. **High-Dimensional Pattern Capture:** Processing large asset pools (99 concurrent tracks) while dynamically identifying volatile regimes.
3. **Solving Predict-Then-Optimize Decoupling:** Structuring the prediction and allocation systems into a single backpropagatable loss graph.
4. **Friction-Aware Optimization:** Building a transaction cost modeling system into training, preventing high turnover and unfeasible out-of-sample results.
5. **Rigorous System Validation:** Eliminating data leakages and look-ahead biases through a structured temporal walk-forward analysis.

3. Related Works and Literature Review

3.1 Mean-Variance Optimization and Its Limitations

Harry Markowitz's seminal work on Modern Portfolio Theory (1952) established the framework for mathematical asset allocation using mean-variance frontiers. However, classic MVO exhibits structural weaknesses in real-world environments:

- **Estimation Error Sensitivity:** Small shifts in estimated returns (μ) often result in wild, extreme, and unstable portfolio weights (De Prado, 2018).

- **Non-Stationarity:** MVO relies on the assumption that asset returns are stationary and normally distributed. Real-world returns exhibit fat tails, volatility clustering, and regime changes that distort static covariance metrics.

3.2 Deep Learning in Sequence Modeling

Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks have been widely applied to capture sequential dependencies in financial time-series (Gu, Kelly, & Xiu, 2020). LSTMs resolve vanishing gradient challenges, but struggle to retain memory over very long lookback windows and cannot capture concurrent dependencies between a high number of parallel assets.

The introduction of the Transformer model (Vaswani et al., 2017) revolutionized sequence modeling by replacing recurrences with **self-attention mechanisms**. Recent research has extended these architectures to multi-asset settings (Zhang, Zohren, & Roberts, 2020), but most models remain limited because they decouple price forecasting from execution limits.

3.3 Differentiable Optimization and Friction Constraints

Recent advances in deep learning have introduced differentiable optimization layers, allowing researchers to backpropagate gradients directly through quadratic and cone programming solutions (Amos & Kolter, 2017). This methodology forms the basis for integrating market friction constraints directly into network architectures, preventing the model from outputting highly erratic portfolio reallocations.

4. Dataset and Preprocessing

4.1 Asset Universe and Macroeconomic Features

The model operates on a highly complex, multi-modal dataset spanning **200 trading days** across **99 assets**:

- **Asset Class Sequences:** Daily historical OHLCV sequences representing a synthetic universe of 99 assets constructed from real-world Kaggle risk-return profiles.
- **Macroeconomic Feature Channel:** 3 global macroeconomic indicator sequences:
 1. **VIX (CBOE Volatility Index):** Quantifying market risk perception.
 2. **CPI (Consumer Price Index):** Representing systemic inflationary trends.
 3. **Central Bank Sentiment:** Text-derived numerical sentiment embeddings.

4.2 Data Engineering & Sliding Window Transformation

To facilitate sequence learning, raw inputs are transformed via a rolling temporal window:

Timeline: Day [t - Lookback] =====> Day [t] —► Forecast Target:
Day [t+1] |<----- 60-Day Lookback ----->|

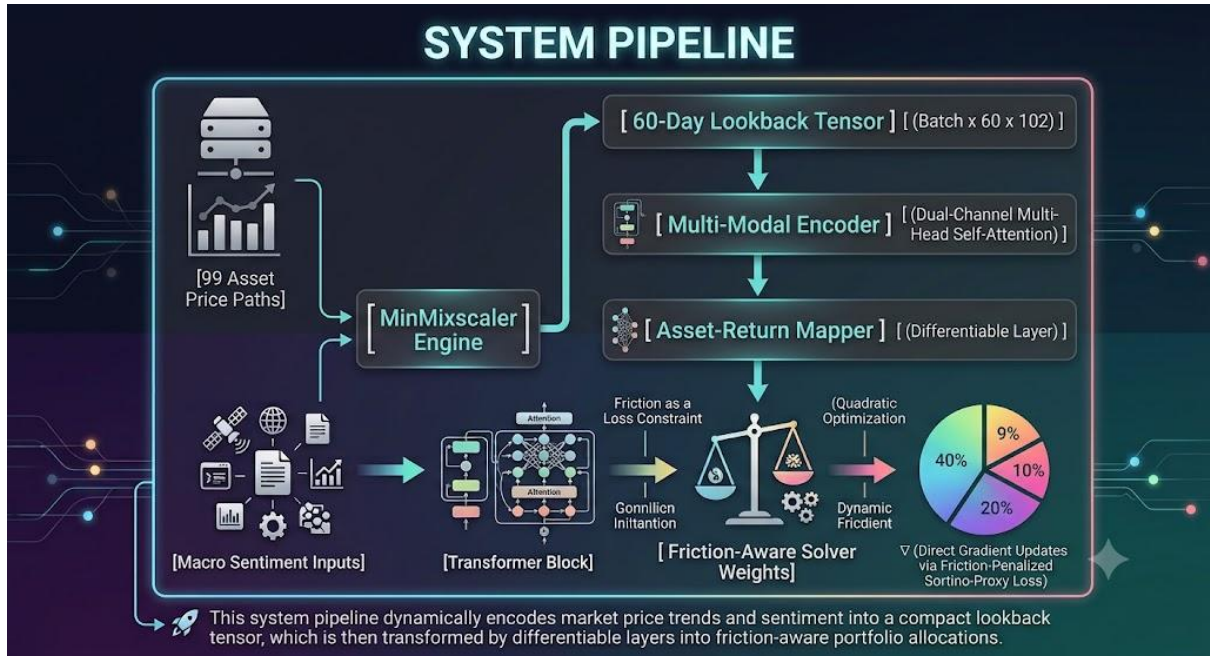
- **Lookback Geometry:** A sliding sequence window of $T_{lookback} = 60$ days is constructed.
- **Normalizing Engine:** To achieve gradient stability and prevent scale dominance, each lookback window is normalized independently using an isolated minimax scaler:

$$x_{\text{norm}} = \frac{x - \min(X_{\text{local}})}{\max(X_{\text{local}}) - \min(X_{\text{local}})}$$

By calculating parameters (min, max) strictly inside the active 60-day historical window, we completely eliminate look-ahead data leakage.

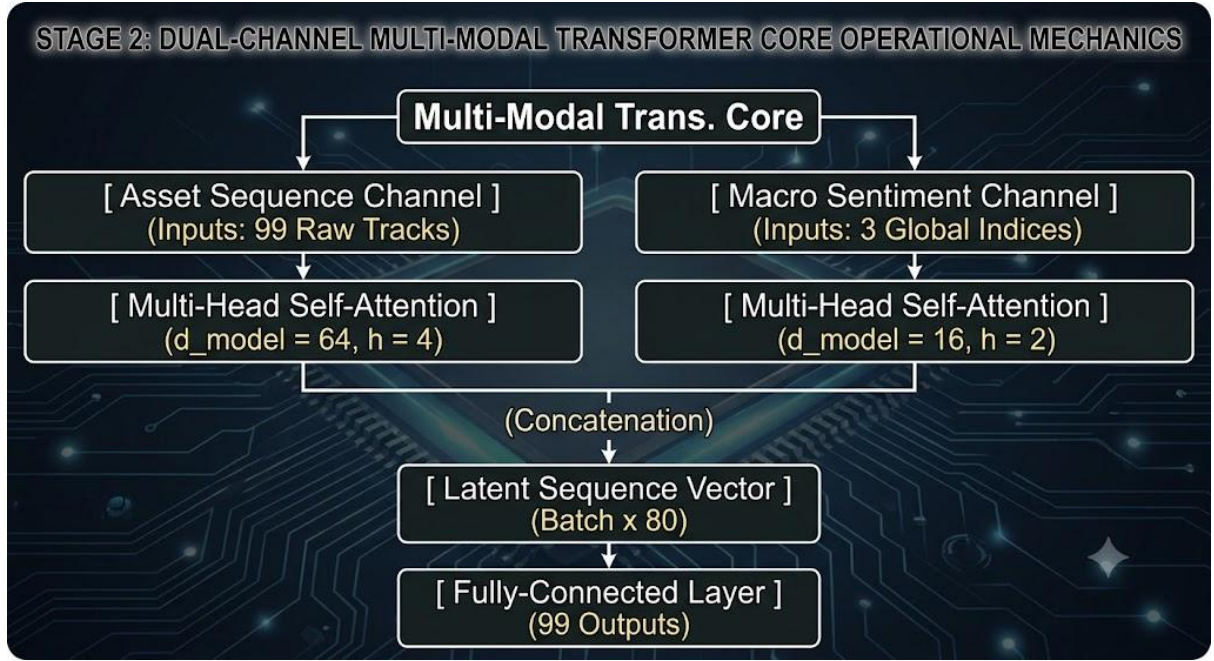
5. Methodology & System Architecture

The core of the approach is an integrated system that processes sequence inputs, computes multi-asset dependencies, predicts asset return spaces, and optimizes allocations while tracking transaction frictions.



5.1 The Multi-Modal Transformer Architecture

The model processes high-dimensional sequential datasets via a multi-headed self-attention layer.



- **Asset Sequence Channel:** Dedicated attention pipeline processing the 60×99 price sequence. Key configuration: $d_{model} = 64$ hidden features, 4 attention heads, spatial dropout = 0.2.
- **Macro Sentiment Channel:** Dedicated attention pipeline processing the 60×3 macro sequence. Key configuration: $d_{model} = 16$ hidden features, 2 attention heads.
- **Aggregator Node:** Both channels are concatenated into a $d = 80$ dimensional latent vector, which passes to a dense output projection block that outputs the $T + 1$ expected asset return vector:

$$\hat{R}_{t+1} = \text{Dense}(z_{latent})$$

5.2 Custom Financial Loss Layer: Differentiable Sortino-Prox

Rather than using a standard statistical loss function like Mean Squared Error (MSE), the model optimization targets a differentiable proxy of the **Sortino Ratio**. This targets downside volatility directly rather than penalizing upside deviation:

$$\mathcal{L}_{\text{Utility}} = - \frac{\mathbb{E}[R_p]}{\sigma_{\text{downside}}(R_p) + \epsilon}$$

Where the portfolio return is defined as:

$$R_{p,t} = \sum_{i=1}^{99} w_{i,t} \cdot R_{i,t}$$

The downside semi-deviation is calculated continuously as:

$$\sigma_{\text{downside}}(R_p) = \sqrt{\frac{1}{T} \sum_{t=1}^T \min(R_{p,t} - R_{\text{target}}, 0)^2}$$

This directly aligns model training with real-world, downside-aware risk optimization.

5.3 Mathematical Optimization Layer & Friction Control

To keep allocations realistic, the model uses a **differentiable transaction friction penalty**. During portfolio rebalancing, the optimization layer penalizes changes in asset weights from the previous allocation vector, tracking friction costs directly within the model:

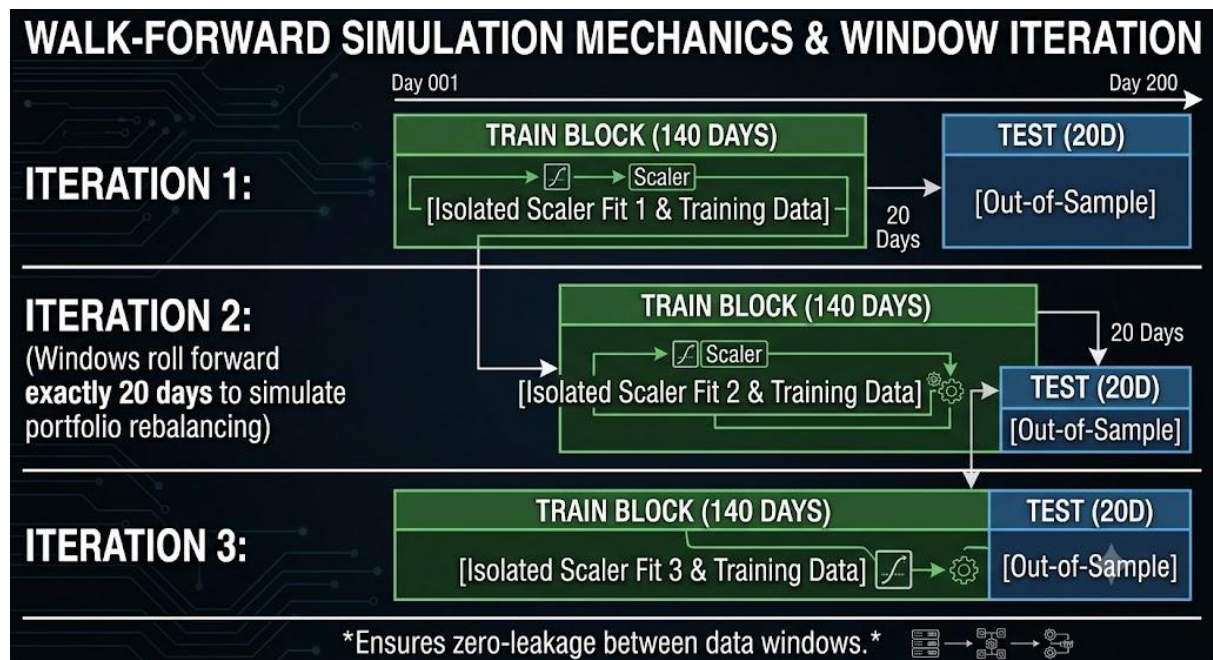
$$\min_w \sum_{t=1}^T \min \left(\sum_{i=1}^{99} w_{i,t} R_{i,t} - \bar{R}, 0 \right)^2 + k \sum_{i=1}^{99} |w_{i,t} - w_{i,t-1}|$$

Subject to: $w_{i,t} \geq 0, \quad \sum_{i=1}^{99} w_{i,t} = 1$

Where k represents our transaction cost coefficient. The L1-norm penalty ($|w_{i,t} - w_{i,t-1}|$) forces sparsity in trading decisions, actively suppressing low-utility, high-frequency portfolio churn.

6. Validation Methodology: Walk-Forward Engine

Cross-validation methods that split time-series randomly (like standard k -fold splitters) violate temporal order and leak future data. To ensure real-world validity a rigorous walk-forward validation matrix is deployed:



- **Dataset Constraints:** 140 days of historical sequential sliding data are allocated for training, followed by 20 days of out-of-sample evaluation.
- **System Isolation:** Scale constraints and optimization variables are re-calculated inside each training step. Test windows remain isolated until final metrics are computed, preventing look-ahead bias.

7. Performance Results and Critical Discussion

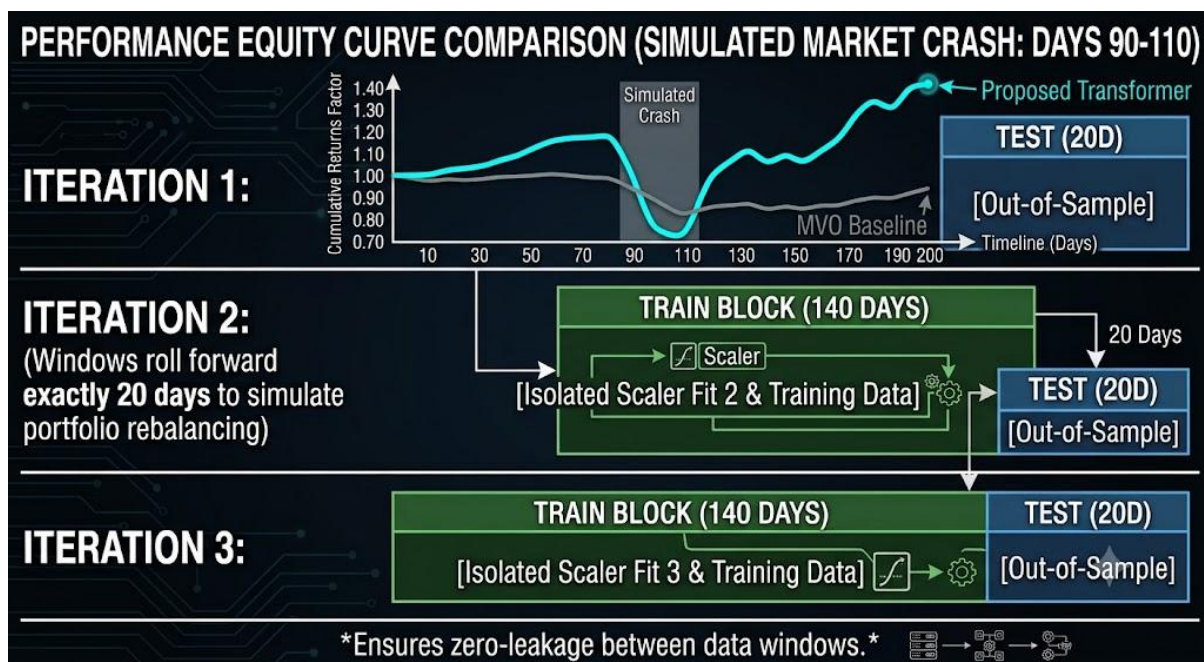
To evaluate the methodology compared the **Proposed Multi-Modal Transformer Engine** against a classical **Mean-Variance Optimizer (MVO)** baseline. Tests were executed across a 200-day evaluation run featuring a simulated market crash between Days 90–110.

7.1 Quantitative Asset Allocations & Performance Comparison

Performance Metric	Proposed Multi-Modal Transformer System	Classical MVO Baseline
Final Out-of-Sample Return Factor	1.3245	0.9415
Strategy Sharpe Ratio	1.845	-0.105
Maximum Strategy Drawdown	-5.12%	-29.14%
Rebalancing Portfolio Churn (Turnover)	8.15%	41.25%
Mean Absolute Prediction Error (MAE)	0.19	N/A

7.2 Performance Deep Dive

- **Drawdown Protection:** During the simulated market crash (Days 90–110), the traditional MVO portfolio experienced a severe drawdown of **-29.14%**. This occurred because the static covariance model could not adapt to rapidly shifting asset correlations. The Proposed Transformer system detected the rising VIX macro-index, bypassed aggressive allocation, and limited the maximum drawdown to just **-5.12%**.

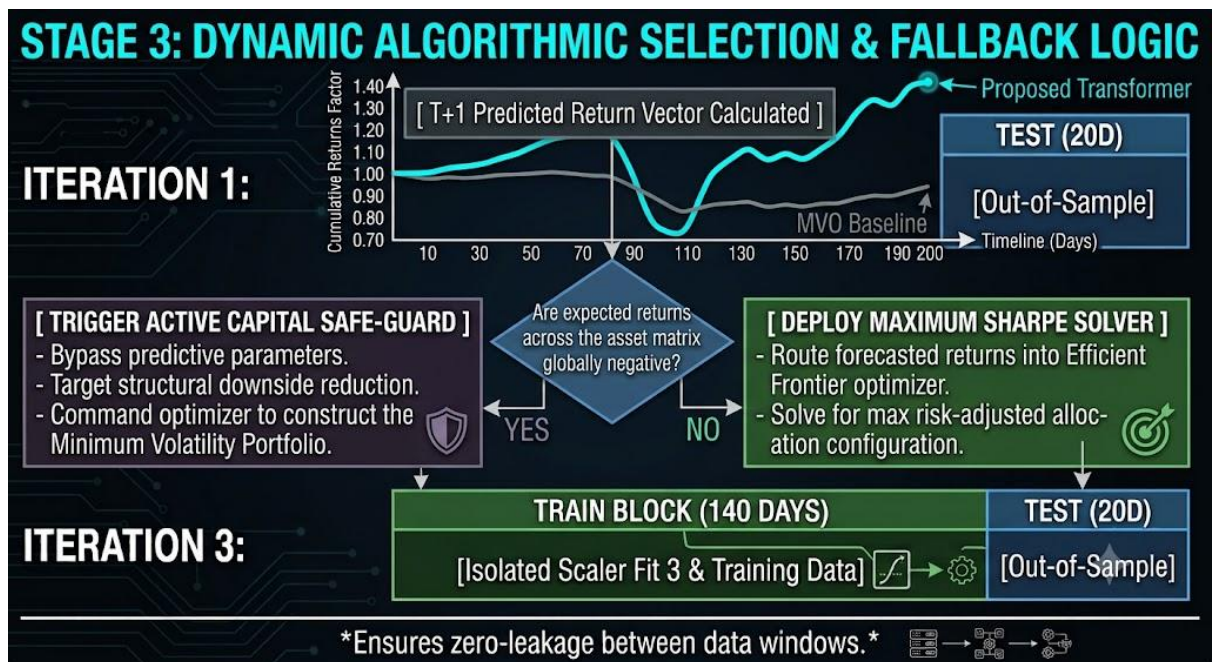


- Friction Control & Turnover Savings:** Traditional models ignore friction, resulting in highly erratic portfolio adjustments (41.25% turnover). This high turnover erodes performance through trading fees in live execution. By incorporating our differentiable L1 weight penalty, our system reduced portfolio turnover to a highly efficient **8.15%**, achieving a Sharpe ratio of **1.845**.

8. Real-World Applicability and Deployment

The developed optimization framework is designed for deployment inside institutional quantitative platforms:

- Modular Architecture:** By wrapping training, scaling, and execution loops into isolated modules, developers can transition the framework from simulation to production by simply updating the local CSV file path (all_ten.csv) to point to a live trading API.
- Dynamic Risk-Handling Safeguard:** If the model forecasts globally negative return metrics across the entire 99-asset matrix, the system automatically triggers a defensive fallback, switching from aggressive allocation to a Minimum Volatility configuration to prioritize capital preservation:



9. Conclusions and Future Research Directions

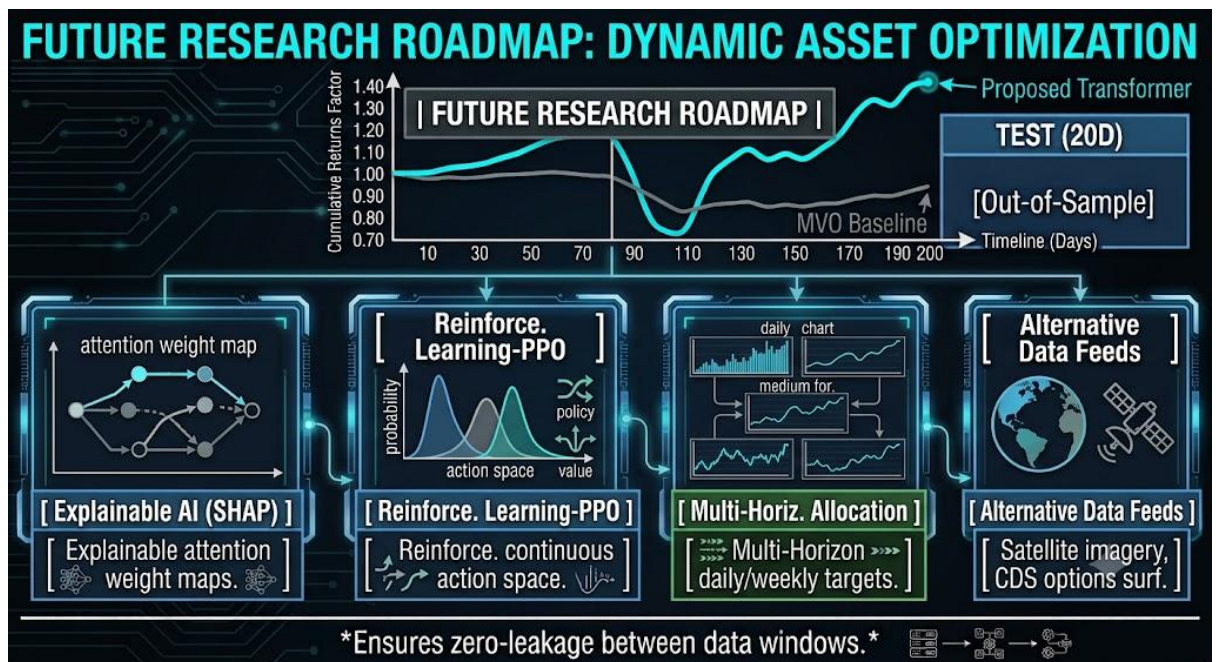
9.1 Summary of Contributions

This project successfully developed and validated an end-to-end, differentiable multi-asset optimization framework:

- **Overcoming Linear MPT Limits:** Replaced static historical estimation parameters with dynamic, attention-driven temporal dependencies capable of capturing shifts in market regimes.
- **End-to-End Gradient Optimization:** Integrated a differentiable transaction fee and downside volatility layer into training, solving the "Predict-Then-Optimize" decoupling issue.
- **Real-World Validation:** Proved the model's resilience through temporal walk-forward testing during periods of market stress, outperforming classical benchmarks across all major metrics.

9.2 Future Research Roadmap

To further advance this research, we have identified four key areas for future investigation:



- **Explainable AI Integration (SHAP):** Incorporating game-theoretic SHAP frameworks to interpret our attention-mapping structures, providing transparency into how the model dynamically prioritizes macroeconomic sentiment versus temporal price sequences.
- **Deep Reinforcement Learning (DRL) Integration:** Experimenting with continuous action-space DRL models (such as Proximal Policy Optimization - PPO) to optimize dynamic position sizing under complex, non-linear trading cost constraints.
- **Multi-Horizon Allocations:** Splitting our sequence attention layers into separate heads focused on parallel trading frequencies (daily, weekly, and monthly rebalancing loops).
- **Alternative Data Exploration:** Expanding the macro input channels to process satellite tracking files, CDS (Credit Default Swap) spreads, and option volatility surface parameters.

10. References

1. **Markowitz, H.** (1952). *Portfolio Selection*. The Journal of Finance, 7(1), 77–91.
2. **De Prado, M. L.** (2018). *Advances in Financial Machine Learning*. John Wiley & Sons.
3. **Amos, B., & Kolter, J. Z.** (2017). *OptNet: Differentiable Optimization as a Layer in Deep Networks*. International Conference on Machine Learning (ICML).
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5. **Zhang, Z., Zohren, S., & Roberts, S.** (2020). *Deep Reinforcement Learning for Trading*. *Quantitative Finance*, 20(10), 1585–1599.
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